Aaryan Kapoor

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Summary

Third-year Computer Science student and first-author undergraduate researcher on multi-step LLM agent pipelines for embedded code generation. Ships AI agents, agent infrastructure, and retrieval systems end to end, backs model claims with held-out benchmarks, and won Best Student Team at the Anthropic Halifax Hackathon.

Technical Skills

Programming: Python, TypeScript, JavaScript, Java, Rust, C, SQL
Agentic AI: LangGraph, Deep Agents, ReAct Agents, Multi-Agent Systems, MCP, DSPy, Agentic RAG, CrewAI
Evals: Held-out Benchmarks, Hidden-Test Oracles, Calibration, Abstention Scoring, Mutation Testing, LangSmith
LLM Systems: llama.cpp, GGUF Quantization, Hybrid Retrieval (RRF), Agent Guardrails, Human-in-the-Loop
Frontend & Backend: React.js, Next.js, Node.js, Express.js, FastAPI, Spring Boot, axum, Tailwind CSS, REST, gRPC
Databases & Tools: PostgreSQL, SQLite, MongoDB, Supabase, Neo4j, Kafka, Docker, Kubernetes, Git, GitHub Actions

Experience

Undergraduate Research – Embedded LLM Research Harness 2026 July – Present
First Author & Lead Researcher, University of New Brunswick | Supervisor: Dr. Andrew Fisher *Fredericton, NB*

- Leading a controlled study of whether a planner, coder, and checker agent pipeline lets a local 9–14B open-weight LLM beat single-prompt generation at writing Arduino firmware, across 5 setups and 46 hand-built tasks. Every setup gets the same evidence and attempt limit, so the result measures structure, not extra feedback.
- Built a 59K-line Python harness (900+ tests, CI) that grades firmware by pin-level Wokwi simulation against hidden checks, so printing `LED ON` without driving the pin fails; canary programs and mutation testing validate the grader.
- Made every run reproducible: the model must return byte-identical output for the same fixed-seed request or it is rejected, with SHA-256-pinned weights, hash-locked dependencies, and a freeze gate on tasks and prompts.

GirlScript Summer of Code 2026
Open Source Contributor *Remote*

- Resolved issues and landed merged PRs across community repositories in a large open-source mentorship program.

CodeSocial 2025
Open Source Contributor *Remote*

- Contributed to open-source projects through the CodeSocial program, fixing issues and getting pull requests merged.

Projects

loopd | *Rust, tokio, axum, rusqlite, TypeScript* | GitHub *June 2026*

- Built a vendor-neutral control plane for AI agent loops: one Rust daemon and TUI that ingests runs from Claude Code, Codex, and SDK agents, and governs them with budget, repeated-action, error-streak, and no-progress policies.
- Published a TypeScript SDK to npm with wire types generated from Rust via ts-rs; covered by 109 unit and 2 integration tests in CI.

Dolos | *Java 21, Spring Boot, Kafka Streams, Neo4j, Spring AI* | GitHub *June 2026*

- Built a real-time financial-crime detection platform (a student-scale Verafin) of 14 Spring Boot services: Kafka Streams and Drools scoring (under 5 μs per transaction), Neo4j fraud-ring detection, and event-sourced case workflow.
- Added a Spring AI copilot exposed as an MCP server; CI deploys the Helm chart to a real k3d Kubernetes cluster and smoke-tests it on every merge to main.

PromptGod | *Manifest V3, TypeScript, Vite* | GitHub · Chrome Web Store *November 2025 – Present*

- Published a Manifest V3 Chrome extension that rewrites prompts inside ChatGPT, Claude, Gemini, and Perplexity, and inserts the result back into the chat box.
- Built local validation, repair, and retry logic that catches weak rewrites, near-echoes, dropped instructions, and wrapper-style outputs before they reach the page; backed by 279 unit, integration, and regression tests.

Software Maintenance Agent | *Python, DSPy, Gemini API, pytest, SQLite* | GitHub *April 2026 – Present*

- Built a local coding agent for small, testable fixes: it takes a task or GitHub issue, reproduces the failure in a sandboxed copy, finds likely files, applies a focused patch, reruns tests, and opens a draft PR with a report.
- Ranks fix locations with hybrid retrieval (BM25 plus embeddings via reciprocal rank fusion) and a JEPA-inspired ranker, logging each step in SQLite with secret redaction, command allow-listing, and path-scoped edits.

ShoreCheck | *Next.js, React, Tailwind, Sentinel-2, Gemini API* | GitHub · Live *September 2026*

- Built a Halifax beach water-quality dashboard pulling lab samples, ECCC weather data, and Sentinel-2 satellite imagery of 43 lakes into one live view; awarded Best Student Team.
- Backtested the rain-based bloom prediction model against 647 satellite readings and reported it as statistically insignificant ($p = 0.49$) instead of overselling accuracy.

Keepline | *Python, FastAPI, Next.js, SQLite, Claude API, Snowflake Cortex* | GitHub · Live *September 2026*

- Built a company-memory engine that turns Slack, email, and ticket history into time-versioned, cited facts, scores single-person knowledge risk, and writes handoff packs and day-one onboarding briefs.
- Its answer agent cites, abstains, or routes to the right expert: on a synthetic 216-question held-out benchmark it cut confidently wrong answers from 71.8% to 22.2% vs. plain search and routed 21 of 21 expert questions correctly (vs. 12).
- Runs natively in Snowflake (Cortex Search, Cortex Agent, Snowpark), so data never leaves the warehouse.

whis | *Python, faster-whisper, Playwright, Windows UI Automation, Retell* | GitHub *September 2026*

- Built solo in 36 hours: a voice agent that runs a Windows 11 PC from plain speech and answers a real phone number.
- Acts mid-sentence (p50 ∼170 ms over 1,350 calls) and asks for a spoken yes before saving, closing, or submitting.

Hackathons

Anthropic Halifax Hackathon – Best Student Team Award *September 2026*

- Won Best Student Team with **ShoreCheck**, a live beach water-quality dashboard for Halifax.

HackAtlantic 2026 *September 2026*

- Built **Keepline**, a cited company-memory tool that flags knowledge held by only one person.

Google Developer Groups Hackathon – Finalist *September 2025*

- Reached the finals with **MedBuddy**, a voice-enabled medication assistant built with Next.js, Express, and Groq.

Dual-Use Technology Hackathon, UNB *May 2026*

- Built **Fluo** in 48 hours: a defence procurement workflow with vendor risk checks and a blockchain audit ledger.

Fredericton Ideation Boost Camp 2026 *January 2026*

- Designed and pitched **Auctus**, an AI funding-discovery platform, and shipped a working MVP in 48 hours.

Certifications & Volunteering

- **Oracle GenAI Certified** and **Oracle APEX Cloud Developer Certified Professional** – Oracle (November 2025)
- **AI Engineer Agentic Track: The Complete Agent & MCP Course** – Ed Donner, Udemy (2026): Built multi-agent systems with the OpenAI Agents SDK, CrewAI, LangGraph, AutoGen, and MCP.
- **RAG Bootcamp** – Udemy (May 2026): Practiced traditional, multimodal, and agentic RAG with LangChain, LangGraph, and LangSmith.
- **Atlantic AI Summit 2026** – Volunteer (June 2026, Fredericton, NB): Supported on-site logistics at Atlantic Canada's applied-AI summit hosted by UNB's RIDSAI.

Education

University of New Brunswick September 2024 – April 2028

Bachelor of Science in Computer Science (3rd Year) *Fredericton, NB*

- **Relevant Coursework**: Data Structures & Algorithms, Computer Networks, Computer Architecture & Machine Language, Python, Java, C, SQL, Statistics, Data Analysis